



Himalayan Glacial Disaster — DRR Concept Diagram

Support options drawing on Japanese technology, following the 25–26 August 2026 Bhote Koshi–Trishuli flood in Nepal (working draft)

Drafted 2026-08-30 · Revised 2026-09-05 · Figures are NDRRMA as of 2026-09-01; search ongoing · Not an official position of any agency

There was no way to know this collapse was coming.
What is in question is what happened after the mountain moved.
Downstream there were 6–10 hours. They did not turn into evacuation.

1 What happened

	08:37	The north face of Langtang Lirung falls, bedrock and all (0.2 km² of rock and ice, 1,200 m drop). It briefly dams the Lhende , then breaches as one surge. M5.2 / Mw 5.7 came from the collapse
	08:50 09:20	Stage gauges stop transmitting (= destroyed). 09:05 DHM becomes aware, 16 min after the wave crossed the border
	09:15	Mass SMS alert: 679,295 messages . Only 4 districts
	11:45	A second collapse on the same slope (M4.2). Search teams are entering the valley — no warning issued
	15:20 16:00	Reaches Devghat → peak 6.57 m (no second peak)

About 10 hours from collapse to the lowest reach · 1,114 dead nationwide, ~4,500 missing

This chronology took a while to settle. First USGS report → IRDR (previous-day collapse + long impoundment) → satellite analysis (no dam), before settling on **ICIMOD / USGS: a brief blockage and breach**. With no instruments upstream, everything rested on remote inference. A GLOF struck the same corridor in July 2025 (11 dead).

2 Diagnosis — most bodies were found where there had been time to escape

District (upstream → downstream)	Margin after alert	Bodies recovered
Rasuwa	-25 min	27
Nuwakot	+5 min	95
Dhading	~1.5–2 h	55
Gorkha	~3–4 h	65
Tanahun	~4–5 h	38
Chitwan	6 h 05 min	321
Nawalparasi E not alerted	~7–8 h	216
Nawalparasi W not alerted	~8–9 h	170

A decisive caveat: “bodies recovered” is where a body was found, not where the person died, and recovery is rate-limited by regaining access (from 29 Aug to 1 Sep the upstream counts caught up: Rasuwa 13 → 27, Nuwakot 51 → 95). That is exactly why the **verification survey (A-0) must precede any equipment donation**. The shape still holds: 489 people → **49.5% of this corridor** — were recovered in the four districts that were never alerted; the three lowest districts alone hold **71.6%**; Rasuwa, closest to the source, holds **2.7%**. The **wave did not travel at one speed**: it is put at 120–190 km/h through the gorge and slowed sharply where the valley opens out. The margin exists only in the slow reach — which is where the dead were concentrated.

4 The measures as a whole — laid out along the lead-time axis

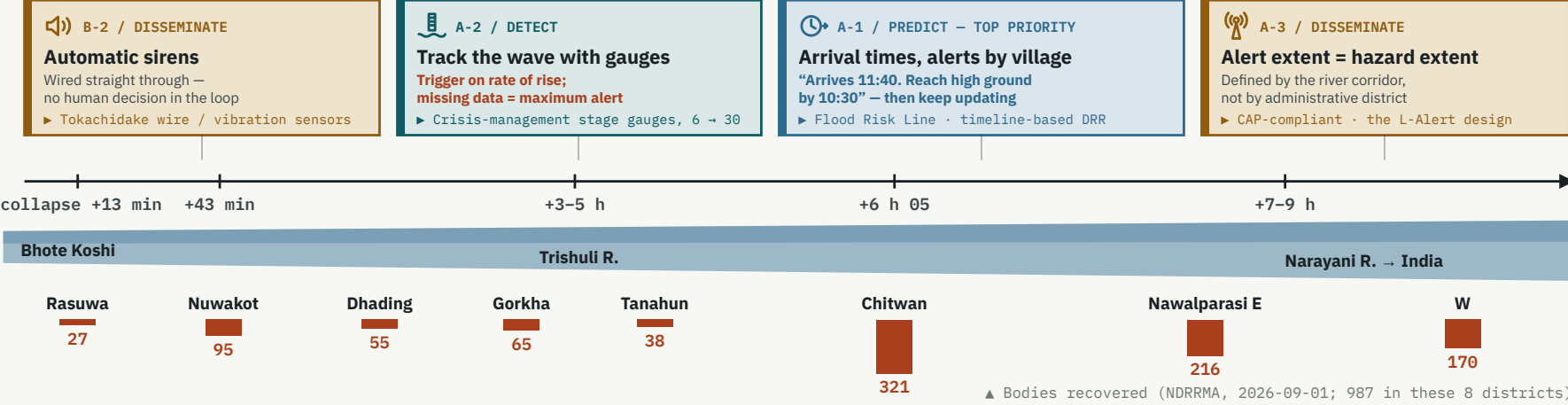
AT THE SOURCE

B-4 / TRANSBOUNDARY Peacetime monitoring Slopes >30°, above 3,000 m, permafrost, near a large river ► ALOS-2/4 · Sentinel Asia	B-1 / DETECT Collapse detection Detects a collapse and alerts automatically, without waiting ► NIED Hi-net · microbarometers	B-3 / PREDICT Blockage assessment The channel was blocked twice here; both needed assessing ► Japan’s emergency-survey rules
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SOURCE MECHANISM — THE READING MOVED, BUT HAS SETTLED

USGS: the 08:37 collapse is the origin → IRDR: a 13:00 collapse the day before, breach after 18–19 h → satellite: no dam Now: bedrock falls with the ice at Langtang Lirung, briefly dams the Lhende, then breaches as one surge		
Orders of magnitude — what fell was rock and ice; the water was picked up en route Frictional melt (even spending all of the 1,200 m drop — 11.8 kJ/kg — caps melt at 3.5% of the ice mass) 18–19 h of channel blockage and impoundment (a catchment the size of the Lhende) Total volume past Devghat (includes baseflow; cannot constrain the source). Peak flow 5,850 m³/s ↳ the excess above baseflow — estimates of the fallen mass diverge, 10⁹–10¹⁰ m³; either way the water came from the river	10 ⁹ –10 ⁶ m³ 10 ⁶ m³ 10 ⁸ m³	
The renewed rise on 30 Aug — the same gap, reproduced four days later the barrier lake was largely empty by 30 Aug Flow rose again and NDRRMA alerted. Noticed not by an instrument but by a report that the river’s roar had grown . Search suspended.		

AFTER THE COLLAPSE — 100 KM IN UP TO 10 HOURS (THE ALERT WENT OUT; EVACUATION DID NOT FOLLOW)



BASIN-WIDE, PEACETIME	A-4 Protocols for tunnel workers and foreign nationals (cell broadcast / tunnel safety standards) where and when each death occurred and which alert was received)	C-1 High-ground refuges and escape routes for every settlement	C-3 Hydropower plants as EWS nodes (monitoring obligations secured through loan and insurance conditions)	A-0 Verification survey (settle the blockage duration and water budget; census of
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Left: what happened at the source, as understood on 2026-09-03. Right: propagation of the flood wave after the collapse (the horizontal axis is not to scale). How much margin there was at the source depends on the reading, but **the 6–10 hours downstream were there on any of them**. Only three arrival times are measured — Syabrubesi 08:50, Betrawati 09:20, Devghat 15:20; the rest are interpolated. On the Trishuli, a 9 m rise at Galchchi and 7 m at Malekhu were recorded within 30 minutes. Bodies recovered is not a count by place of death, and recovery is rate-limited by regaining access. **For the first ~20 km there was little water; the flow was debris, taking up river water as it went** — hence sirens upstream, forecasting downstream. **In places it ran 400 m back upstream**: fleeing along the river is not safe.

5 Japanese technology and the bodies that would deliver it

DETECT NIED / Japan Meteorological Agency Hi-net, V-net; automatic seismic and infrasound detection of lahars Stations sit inside Nepal, so the system can be built without waiting for agreement with China . It needs no identification of mechanism, so it works while views still diverge.	DETECT MLIT / Instrument manufacturers Crisis-management stage gauges; wire and vibration sensors The “under ¥1 M, five years unpowered, retrofittable” specification adopted after Typhoon Lionrock in 2016 (24 dead in Iwaizumi).
PREDICT PWRI / ICHARM / MLIT Flood Risk Line; timeline-based DRR This disaster itself left measured data with which to calibrate the model .	DELIVER MIC / mobile network operators Emergency cell broadcast (ETWS / CB); L-Alert Broadcasts to every handset in several languages without any subscriber list, and reaches roaming foreign SIMs .
CROSS-BORDER JAXA / Sentinel Asia / ADRC ALOS-2 and -4 (L-band SAR); International Disasters Charter The source was inside Nepal , so China’s warning system did not cover it. There is distinct value in Japan observing and distributing neutrally, as a third country .	STRUCTURAL MLIT Sabo Dept. / JICA / JBIC and NEXI Emergency survey of landslide dams; unmanned construction A statutory procedure that runs impounded volume → breach discharge → inundation estimate within 24–48 h . Needed both for the brief initial blockage and for the barrier lake that overtopped on 28 Aug.

6 Sequence of implementation

WHEN	WHAT	WHO
0–3 mo	A-0 verification survey — settle the blockage duration and the water budget; census of place and time of death and of which alert was received. In parallel, satellite monitoring of the unstable mass	Joint academic survey team / JAXA / ADRC
0–1 yr	A-1 arrival-time forecasting; A-3 review of alert coverage; A-4 tunnel and foreign-national protocols	ICHARM / JICA / MIC
1–2 yr	B-1 source detection network (detects an upstream collapse and alerts automatically); B-2 automatic sirens; B-3 transfer of the channel-blockage procedure. Stage gauges to 30+	NIED / MLIT Sabo Dept.
2–5 yr	C-1 high-ground refuges and escape routes; C-2 land-use guidance; C-3 power plants as nodes, with financial conditions	JICA / JBIC and NEXI
ongoing	Build the transboundary data-sharing framework; institutionalise Japan’s observation supply as a neutral third party	MOFA / ICIMOD / ADRC

On equipment dying within three years: require no mains power (five-year battery); use no consumables; duplicate the link (satellite IoT); hold spares and trained repairers locally; and pay against **95%+ annual uptime** rather than against delivery.

7 Outside the scope

Catching this collapse in advance from orbit is hard Rock and ice breaking at once leaves little movement to see beforehand; the ground-temperature anomaly reported two days earlier was found afterwards. Nepali experts say the same — that this was an event no early warning system could have caught . What this diagram addresses is the downstream reach, after the mountain moved.
This is beyond what check dams can hold Catching a debris flow that runs more than 100 km with a structure is not realistic. What works is high-ground refuges, training dykes on tributaries, and designing rebuilt structures so they are not washed away.
Rainfall data could not see this one No rain was involved in this flood. It seems better to keep separate detection paths for rain-driven and glacier-driven floods, and share only how the warning reaches people downstream .
DC Report cannot be the primary path in the mountains Nepal sits near the western edge of the service area, and in deep valleys there will be places where the satellite cannot be seen high enough. Start by measuring, on the ground, where reception actually works .

Legend (what the colours mean)

- Detection & observation
- Prediction & decision
- Dissemination & alerting
- Action & structures
- Cross-border & institutions
- Human losses

Group A = immediate (within 1 yr) / **Group B** = medium term (2 yr; where Japan is distinctively strong) / **Group C** = structures and land use (3–10 yr). Priority = lead time created ÷ cost × capacity to maintain.

Glossary

DHM: Dept. of Hydrology and Meteorology, Nepal
NDRRMA: National Disaster Risk Reduction and Management Authority
GLOF: glacial lake outburst flood
ICIMOD: Int’l Centre for Integrated Mountain Development
ICHARM: Int’l Centre for Water Hazard and Risk Management

CAP: Common Alerting Protocol
ETWS / CB: cell-broadcast emergency alerting
SAR: synthetic aperture radar (sees through cloud)
Unmanned construction: earthworks by remotely operated machines
Timeline DRR: planning backwards from the arrival time



Find it early upstream. Deliver time downstream.

What this disaster lacked was not a means of knowing the danger, but a path from knowing to escaping. On 30 Aug, too, it was a human ear that noticed. Sources: GFZ, ICIMOD, Copernicus / Planet satellite analysis, IRDR, USGS, NDRRMA, MLIT, NIED, JAXA, JICA and others